Lorenzo Zambardi

B.Sc. Student in Mathematical Sciences for Artificial Intelligence (SMIA)

Sapienza University of Rome · Faculty of Information Engineering, Informatics and Statistics

Rome, Italy · Italian Citizen · Born January 19, 2005

lorenzozambardi.github.io · github.com/lorenzozambardi

Profile

Undergraduate student in Mathematical Sciences for Artificial Intelligence (SMIA) at Sapienza University of Rome. Academic focus centers on the formal mathematical foundations of AI, combining theoretical rigor with computational implementation and numerical prototyping.

Education

Sapienza University of Rome — Rome, Italy
B.Sc. in Mathematical Sciences for Artificial Intelligence (SMIA) — **2024 – Present**

- **Expected Graduation**: July 2027
- **Degree Progress**: 126 / 180 ECTS completed (**70.0%** of total program requirements)
- **Academic Standing**:
  - **Weighted Average (GPA)**: **27.95** / 30 (approx. **3.73** / 4.0 US)
  - **Adjusted Weighted Average** (Sapienza graduation basis, lowest 12 ECTS excluded): **28.47** / 30 (approx. **3.80** / 4.0 US)
- **Selected Coursework**:
  - **Linear Algebra and Algebraic Structures** (12 ECTS) – **30 / 30**
  - **Machine Learning** (9 ECTS) – **30 / 30**
  - **Numerical Methods** (9 ECTS) – **30 / 30**
  - **Mathematical Analysis II** (9 ECTS) – **29 / 30**
  - **Algorithms and Complexity** (6 ECTS) – **29 / 30**
- **Prospective Thesis Area**: Mathematical Foundations of Machine Learning & Continuous/Discrete Optimization.

Liceo Scientifico Camillo Cavour — Rome, Italy
High School Diploma (Scientific Curriculum) — **2019 – 2024**

- **Final Grade**: **100 / 100**
- Advanced secondary education in mathematics, physics, and scientific disciplines.

Honors & Academic Awards

Academic Merit Tuition Waiver (*Status Studente Meritevole*) — **2024 – 2026**
Sapienza University of Rome

High School Diploma (100 / 100) — **2024**
Liceo Scientifico Camillo Cavour, Rome

Projects & Computational Implementations

Iron Stats | *Architect & Mathematical Modeler* github.com/lorenzozambardi/Iron-Stats
Computational engine for biomechanical simulation, hypertrophy programming, and multi-objective optimization

- Conceived and formulated the theoretical framework, replacing conventional raw tonnage metrics with continuous biomechanical and physiological modeling.
- Modeled continuous muscle tension across joint Range of Motion (ROM) utilizing **3rd-degree Bézier curves**, capturing non-linear moment arms and dynamic joint lever variations.
- Formulated dynamic physiological simulation metrics: effective repetitions based on High-Threshold Motor Unit (HTMU) recruitment, dynamic Time Under Tension (TUT) incorporating involuntary contraction deceleration, and exponential Central Nervous System (CNS) fatigue curves.
- Formulated and solved a scalarized **multi-objective numerical optimization problem** to synthesize mathematically balanced exercise volume, progression rates, and fatigue mitigation protocols.
- **Technologies**: Python, NumPy, SciPy (numerical optimization routines), Matplotlib.

Neovim for STEM | *Creator & Maintainer* github.com/lorenzozambardi/nvim-for-stem
Modular, high-performance editor engineered for mathematical research and rapid LaTeX authoring

- Architected an ultra-fast, modular development environment inspired by **Gilles Castel's mathematical note-taking paradigm**, engineered to eliminate friction between theoretical derivation and digital typesetting.
- Engineered an extensive real-time LaTeX authoring workflow featuring **context-aware, auto-expanding mathematical snippets** for rapid typing of complex theorems, proofs, matrices, and multi-index summations at lecture speed.
- Integrated an end-to-end Python development toolchain (Pyright LSP, Black code formatting), unified cross-platform POSIX installation scripts, and bidirectional cursor synchronization (SyncTeX) with Zathura PDF viewer.
- **Technologies**: Lua, Neovim API, LaTeX / SyncTeX, Pyright LSP, Bash, Arch Linux.

Technical Skills

- **Programming Languages**: Python (Proficient; numerical modeling, scientific computing, PyTorch ecosystem), Rust (Foundational), SQL (Relational database querying & design), Assembly (x86/ARM architecture fundamentals).
- **Scientific Computing & ML Frameworks**: PyTorch, NumPy, SciPy, Scikit-learn, Matplotlib.
- **Operating Systems & Developer Tools**: Arch Linux (daily driver environment), Neovim / Vim, Git & GitHub (version control, open-source workflow), LaTeX / SyncTeX, Unix Shell / Bash.

Languages

- **Italian**: Native speaker.
- **English**: Fluent.
- **Spanish**: Fair knowledge.